A THREE-PART SERIES · MAY 2026

Bainbridge's Debt

Forty years of HCI and human-factors research already named, documented, and partly solved banking's human-automation failure modes. Banking keeps building governance as if none of it were ever published.

SERIES MAP

One argument, three movements

1 The original ironies

Bainbridge, 1983: better automation makes the human's residual job harder, not easier. Skill atrophy, the training paradox, and the opacity problem.

2 How it shows up in banking oversight

Banking runs AI across the full automation range but applies one 'effective challenge' expectation to all of it. The automation-bias evidence says 'the analyst will catch it' is empirically unreliable.

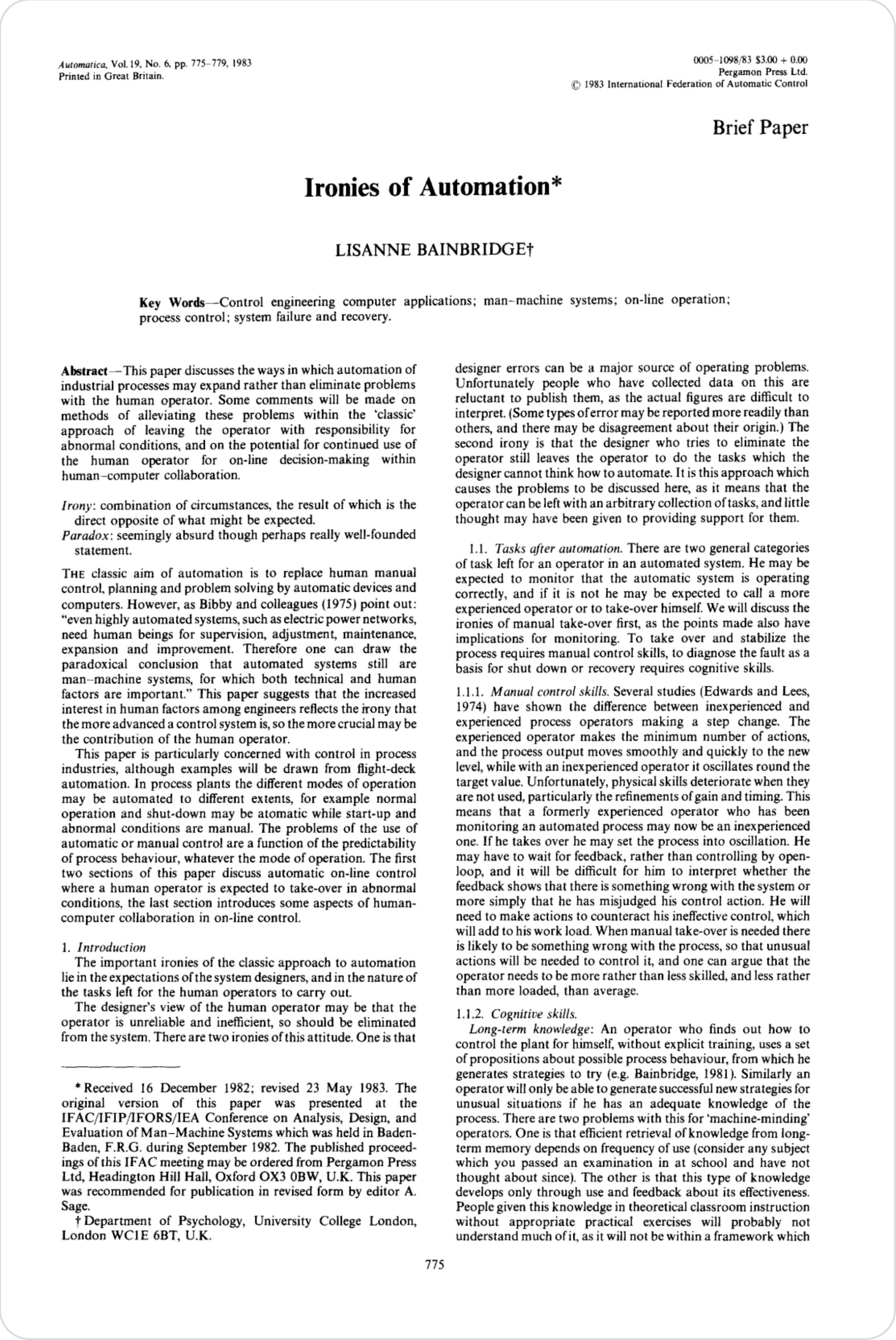
3 The vocabulary mismatch

Two fields solved the same problem for forty years without citing each other. The HCI ↔ banking-governance crosswalk is tractable, overdue, and still doesn't exist.

PART 1 · IRONIES OF AUTOMATION

Six Pages That Keep Coming True

In 1983 Lisanne Bainbridge named every human-automation failure mode banking has since reproduced — across ATMs, credit scoring, RPA, and now agentic AI. The paper has never been cited in US banking supervisory text.


Automatica, Vol. 19, No. 6, pp. 775–779, 1983
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Brief Paper

Ironies of Automation*

LISANNE BAINBRIDGE†

Key Words—Control engineering computer applications; man–machine systems; on-line operation; process control; system failure and recovery.

Abstract—This paper discusses the ways in which automation of industrial processes may expand rather than eliminate problems with the human operator. Some comments will be made on methods of alleviating these problems within the 'classic' approach of leaving the operator with responsibility for abnormal conditions, and on the potential for continued use of the human operator for on-line decision-making within human–computer collaboration.

Irony: combination of circumstances, the result of which is the direct opposite of what might be expected.
Paradox: seemingly absurd though perhaps really well-founded statement.

THE classic aim of automation is to replace human manual control, planning and problem solving by automatic devices and computers. However, as Bibby and colleagues (1975) point out: "even highly automated systems, such as electric power networks, need human beings for supervision, adjustment, maintenance, expansion and improvement. Therefore one can draw the paradoxical conclusion that automated systems still are man–machine systems, for which both technical and human factors are important." This paper suggests that the increased interest in human factors among engineers reflects the irony that the more advanced a control system is, so the more crucial may be the contribution of the human operator.

This paper is particularly concerned with control in process industries, although examples will be drawn from flight-deck automation. In process plants the different modes of operation may be automated to different extents, for example normal operation and shut-down may be atomatic while start-up and abnormal conditions are manual. The problems of the use of automatic or manual control are a function of the predictability of process behaviour, whatever the mode of operation. The first two sections of this paper discuss automatic on-line control where a human operator is expected to take-over in abnormal conditions, the last section introduces some aspects of human-computer collaboration in on-line control.

1. *Introduction*

The important ironies of the classic approach to automation lie in the expectations of the system designers, and in the nature of the tasks left for the human operators to carry out.

The designer's view of the human operator may be that the operator is unreliable and inefficient, so should be eliminated from the system. There are two ironies of this attitude. One is that designer errors can be a major source of operating problems. Unfortunately people who have collected data on this are reluctant to publish them, as the actual figures are difficult to interpret. (Some types of error may be reported more readily than others, and there may be disagreement about their origin.) The second irony is that the designer who tries to eliminate the operator still leaves the operator to do the tasks which the designer cannot think how to automate. It is this approach which causes the problems to be discussed here, as it means that the operator can be left with an arbitrary collection of tasks, and little thought may have been given to providing support for them.

1.1. *Tasks after automation.* There are two general categories of task left for an operator in an automated system. He may be expected to monitor that the automatic system is operating correctly, and if it is not he may be expected to call a more experienced operator or to take-over himself. We will discuss the ironies of manual take-over first, as the points made also have implications for monitoring. To take over and stabilize the process requires manual control skills, to diagnose the fault as a basis for shut down or recovery requires cognitive skills.

1.1.1. *Manual control skills.* Several studies (Edwards and Lees, 1974) have shown the difference between inexperienced and experienced process operators making a step change. The experienced operator makes the minimum number of actions, and the process output moves smoothly and quickly to the new level, while with an inexperienced operator it oscillates round the target value. Unfortunately, physical skills deteriorate when they are not used, particularly the refinements of gain and timing. This means that a formerly experienced operator who has been monitoring an automated process may now be an inexperienced one. If he takes over he may set the process into oscillation. He may have to wait for feedback, rather than controlling by open-loop, and it will be difficult for him to interpret whether the feedback shows that there is something wrong with the system or more simply that he has misjudged his control action. He will need to make actions to counteract his ineffective control, which will add to his work load. When manual take-over is needed there is likely to be something wrong with the process, so that unusual actions will be needed to control it, and one can argue that the operator needs to be more rather than less skilled, and less rather than more loaded, than average.

1.1.2. *Cognitive skills.*

Long-term knowledge: An operator who finds out how to control the plant for himself, without explicit training, uses a set of propositions about possible process behaviour, from which he generates strategies to try (e.g. Bainbridge, 1981). Similarly an operator will only be able to generate successful new strategies for unusual situations if he has an adequate knowledge of the process. There are two problems with this for 'machine-minding' operators. One is that efficient retrieval of knowledge from long-term memory depends on frequency of use (consider any subject which you passed an examination in at school and have not thought about since). The other is that this type of knowledge develops only through use and feedback about its effectiveness. People given this knowledge in theoretical classroom instruction without appropriate practical exercises will probably not understand much of it, as it will not be within a framework which

* Received 16 December 1982; revised 23 May 1983. The original version of this paper was presented at the IFAC/IFIP/IFORS/IEA Conference on Analysis, Design, and Evaluation of Man–Machine Systems which was held in Baden-Baden, F.R.G. during September 1982. The published proceedings of this IFAC meeting may be ordered from Pergamon Press Ltd, Headington Hill Hall, Oxford OX3 0BW, U.K. This paper was recommended for publication in revised form by editor A. Sage.
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Three ironies, one underlying problem

IRONY 01

Skill atrophy

Automation removes the routine practice that builds the skill — yet leaves humans responsible for exactly the exceptional cases that skill is for.

IRONY 02

Training paradox

Operators now need *more* skill than their manual-era predecessors, but rare interventions make that skill impossible to maintain through practice.

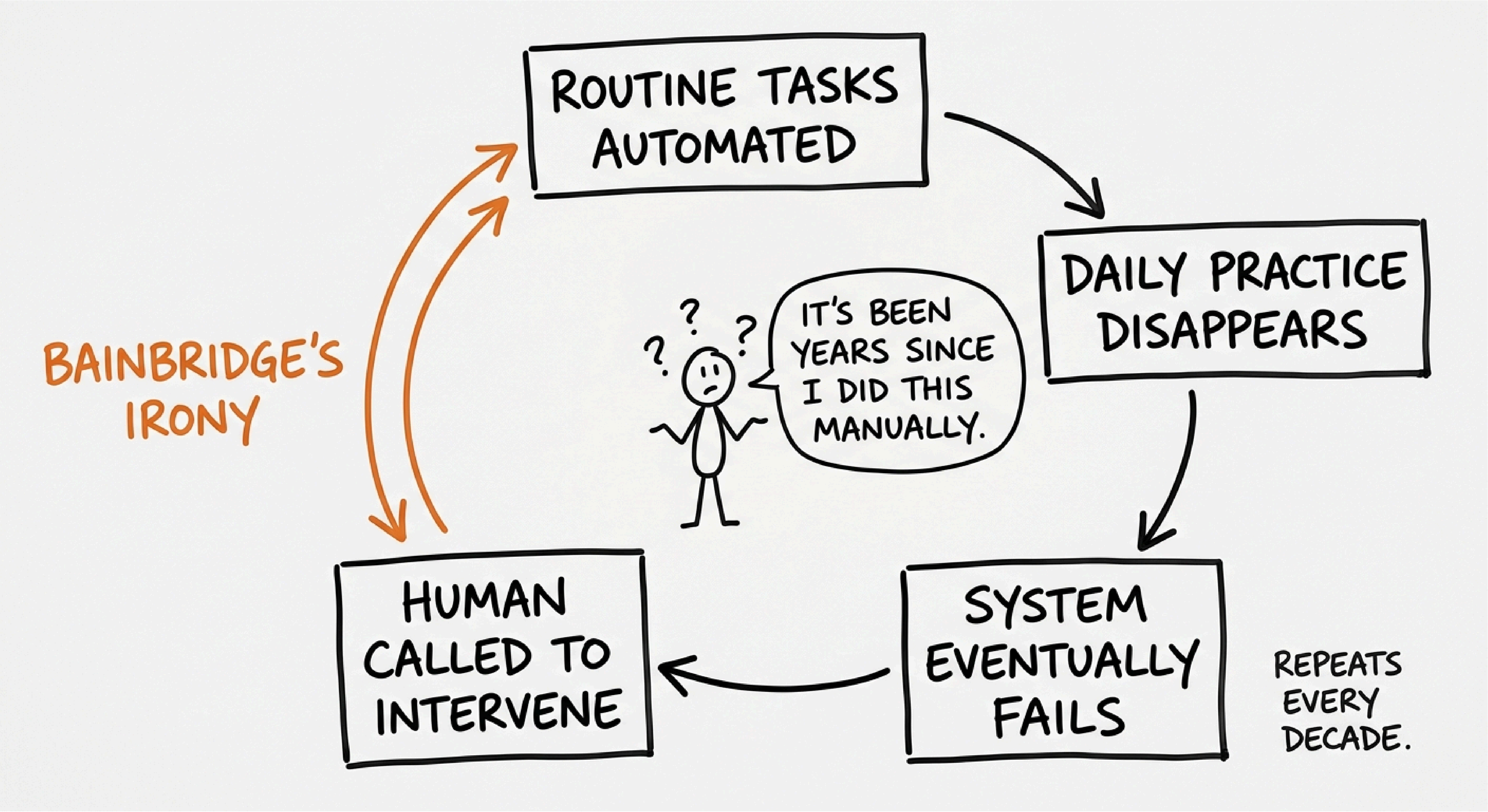
IRONY 03

Opacity

You cannot meaningfully intervene in a system you cannot inspect. A 175-billion-parameter transformer has no legible internal state to oversee.

> Automation creates a human oversight role without designing for the human factors that oversight requires.

Each wave removes the practice it depends on



FOUR WAVES, SAME IRONIES

Banking has run this experiment four times

WAVE	SKILL ATROPHY	TRAINING PARADOX	TRANSPARENCY IRONY
ATM 1969–85	Tellers repositioned to exception work they weren't trained for	Daily transaction practice replaced by an advisory role	Minimal — process states stayed legible
Credit scoring 1985–98	Officers stop forming applicant intuition; the score dominates	Rubber-stamping removes the case-by-case practice	Model internals opaque — 'black box' is the era's term
RPA 2000–18	Analysts inherit the hard 20% without practice on the easy 80%	Exception skill needs the breadth the bots eliminated	Bot logic undocumented; successors inherit blind queues
Agentic AI 2024–	Volume and velocity make exhaustive review infeasible	Few have skills to intervene in a misbehaving pipeline	Chain-of-thought is not decision provenance

WHAT MAKES THE FOURTH WAVE DIFFERENT

The decade of lag is gone

<2 yrs

From research to enterprise deployment — faster than institutional learning cycles.

The first three waves each took roughly a decade for the failure to surface. The agentic wave arrives with all three ironies present *from the moment of deployment* — and SR 26-2 footnote 3 carves generative and agentic AI out of scope entirely.

> Each irony predicts a measurable signal — falling override rates, untestable training, chain-of-thought standing in for provenance. None is currently required.

SHE DIDN'T ONLY NAME THE IRONIES

Three prescriptions, mostly unimplemented

01

Periodic manual engagement

Hands-on control each shift — or high-fidelity simulation. In banking: override drills against real historical failures, not annual policy attestations.

02

Train for adaptive reasoning

Unknown failures can't be drilled. "It is ironic to train operators in following instructions and then put them in the system to provide intelligence."

03

Fail obviously, not gracefully

Systems should fail visibly. An LLM that confidently hallucinates while appearing to reason is the structural inverse of this.

THE GAP, QUANTIFIED

0

times "Ironies of Automation" appears in US banking supervisory text

~3,500 citations in aviation, medicine, and industrial control. Zero in SR letters, OCC bulletins, FDIC FILs, CFPB circulars, or FINRA notices across two decades.

The paper is six pages. It was published in 1983. That's the debt.

PART 2 · AUTOMATION BIAS IN BANKING

"The Analyst Will Catch It"

Banking runs AI across the full automation range but applies one 'effective challenge' expectation to all of it. The human-factors literature has had names for the failure mode at each level for thirty years.

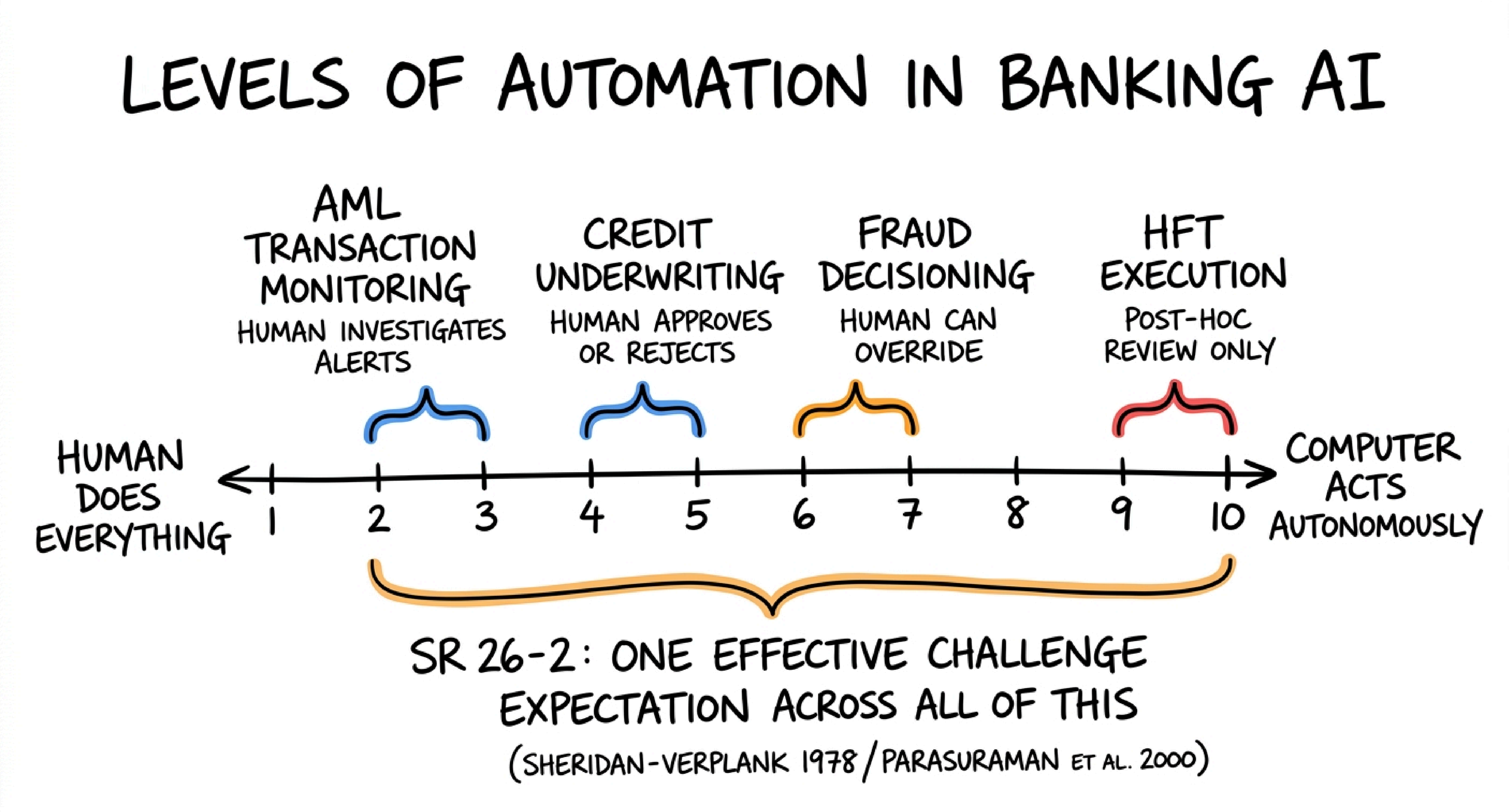
A SCENE YOU'VE SEEN

400 alerts, 96% false positives, one shift

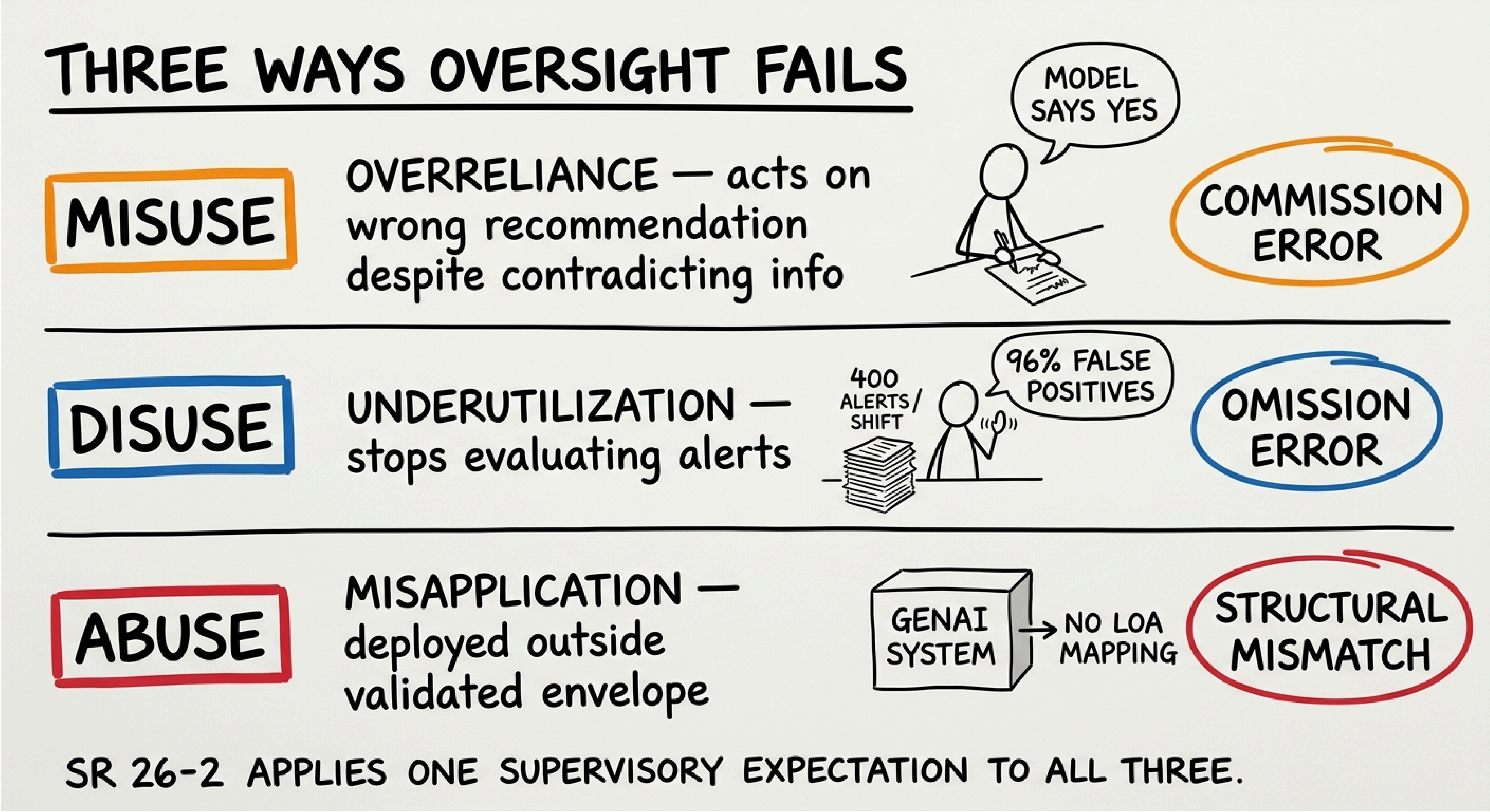
By mid-shift the AML analyst's review is rote dismissal, not investigation. She is performing the motions of oversight without its substance. Parasuraman and Riley named it in 1997: **disuse**.

> Her supervisor doesn't see disuse. They see a human-in-the-loop with an acceptably low false-negative rate.

Banking runs the whole range at once



Three failure modes, three mitigations



"The analyst will catch it" is empirically unreliable

worse

SKITKA ET AL. 1999

Participants with an imperfect aid made *more* errors than those with no aid at all — including omission errors on events the aid never flagged.

21%

BAHNER 2008 / MANZEY 2010

Commission errors even when participants were explicitly warned the aid was fallible and told to cross-check every diagnosis.

≠

BANSAL ET AL. 2021

Explanations alone did not produce complementary performance. Humans accepted or rejected wholesale; case-by-case calibration was rare.

> Automation bias cannot be prevented by training or instructions. Prior failure experience reduced it — but never eliminated it.

"SO JUST REMOVE THE HUMAN"

The Level-10 case — and why banks can't take it

Answer 01

Detection, not just performance

NIST AI 800-4: models may behave when monitored and diverge when not. Removing oversight removes the mechanism that catches misalignment.

Answer 02

Correlated failure at speed

Automation bias is case-level and correctable. A Level-10 failure is the same flaw across every case at once — Knight Capital lost $440M in 45 minutes.

Answer 03

Accountability stays put

SR 26-2, Article 14, and ECOA all presuppose a human who can be held accountable. Removing involvement leaves the accountability on paper.

One framework names the mechanism. One doesn't.

EU AI ACT · ARTICLE 14

Names 'automation bias'

Oversight persons must "remain aware of the tendency of over-relying" on a high-risk system. Credit scoring is explicitly in scope.

✓ automation bias ✓ human oversight ✓ high-risk tiering

US · SR 26-2

Names the structure, not the mechanism

No "automation bias." No "levels of automation," "supervisory control," "skill degradation," or "alert fatigue." GenAI/agentic excluded by footnote 3.

✗ automation bias ✗ levels of automation ✗ alert fatigue

 WARM-UP

Bainbridge's first irony, 'skill atrophy,' claims that automation...

- **A** degrades all human skills through general disuse
- **B** removes the routine practice that builds the skill needed for the exceptional cases humans stay responsible for
- **C** makes operators physically slower at manual tasks over time
- **D** renders human training programs entirely obsolete

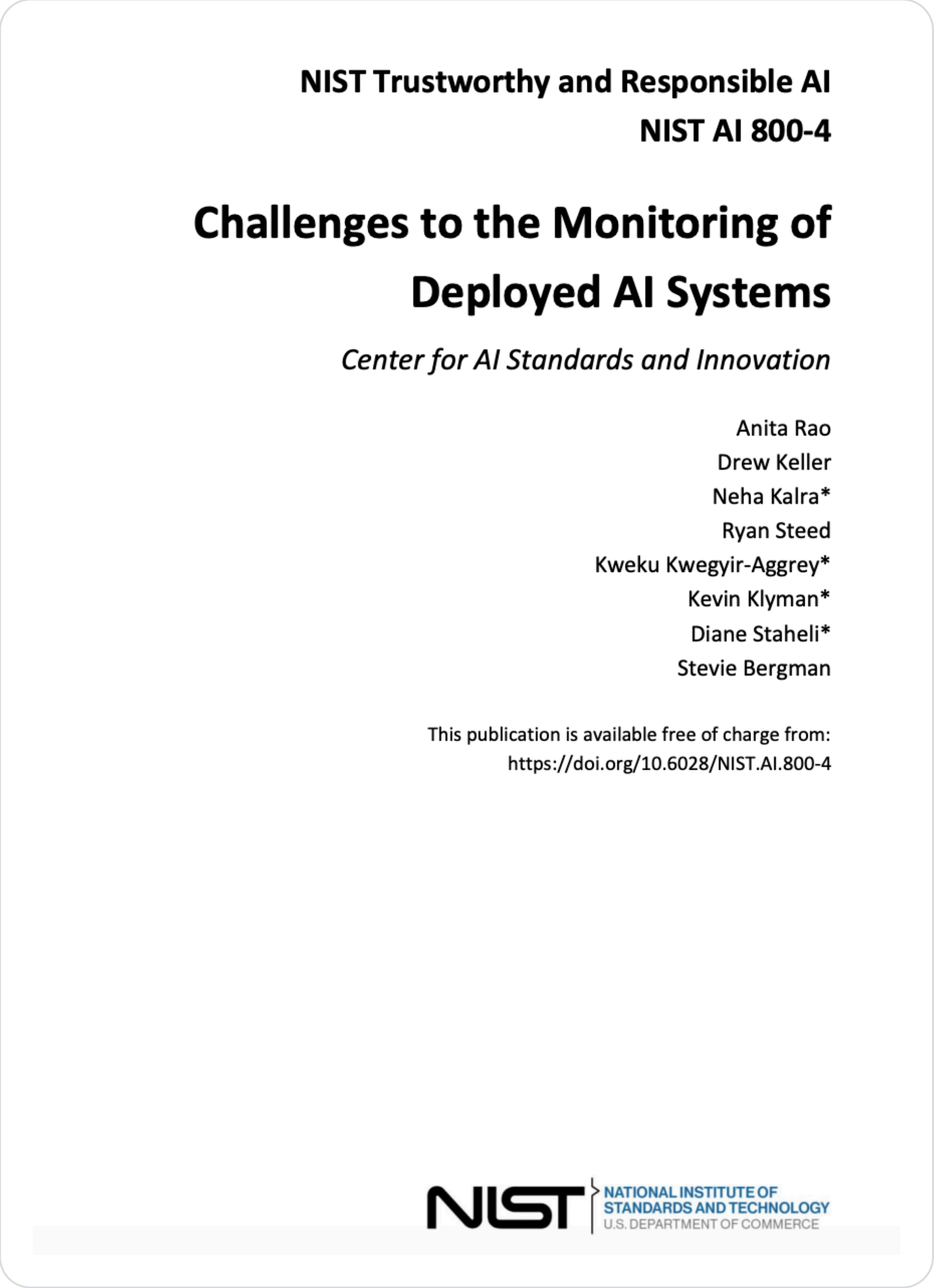
At LOA 2–3 (an AML analyst triaging a 96%-false-positive queue), which failure mode dominates — and why does "effective challenge" miss it?

- **A** Misuse — the analyst over-relies on correct flags, and effective challenge is too strict
- **B** Disuse — chronic false alarms turn review into rote dismissal, which a present-but-disengaged human still satisfies on paper
- **C** Abuse — the system runs outside its validated scope, which effective challenge already covers
- **D** None — Level 2–3 is the safest tier, so effective challenge is sufficient

PART 3 · THE HCI ↔ BANKING CROSSWALK

The Last Mile That No One Owns

Two disciplines spent forty years solving the same problem without once citing each other. The crosswalk between HCI vocabulary and banking governance is tractable — and still doesn't exist.


NIST Trustworthy and Responsible AI
NIST AI 800-4

Challenges to the Monitoring of Deployed AI Systems

Center for AI Standards and Innovation

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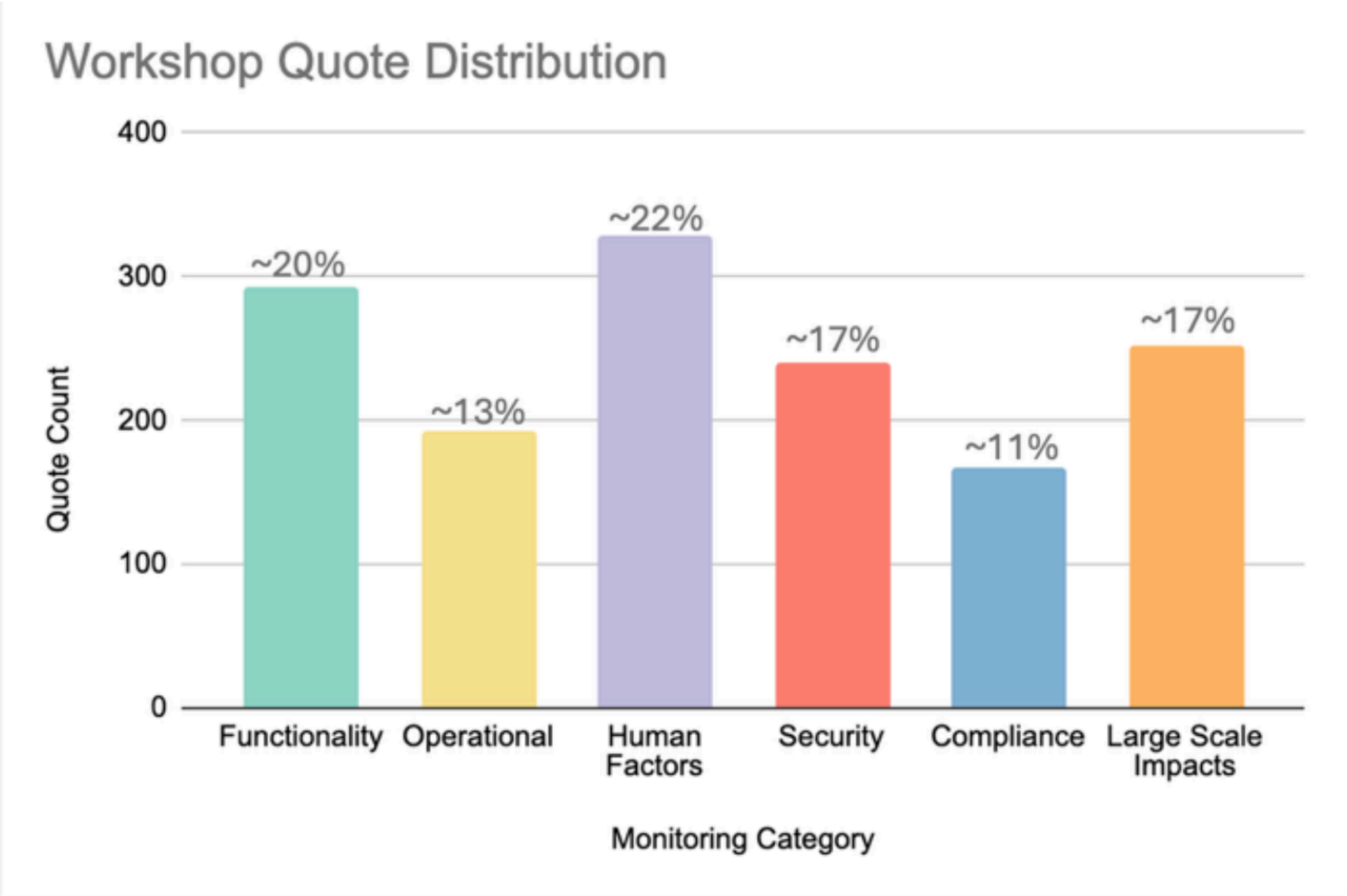
This publication is available free of charge from:
https://doi.org/10.6028/NIST.AI.800-4

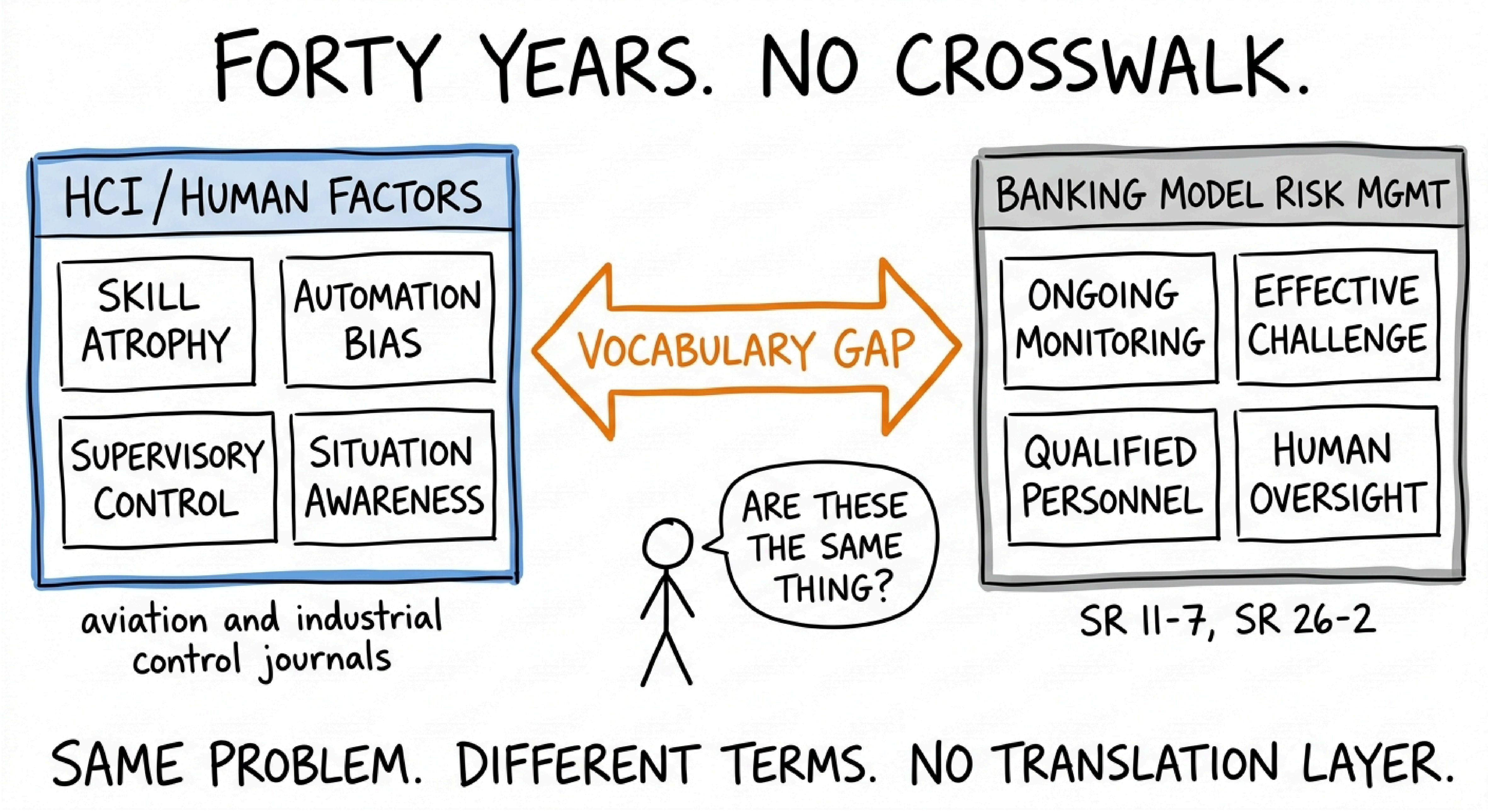
NIST NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
U.S. DEPARTMENT OF COMMERCE

Of six monitoring categories, one is the hardest



Not the most technically demanding. Not the most resource-intensive. The one where practitioners reported the greatest difficulty collecting information — and gauging whether they were doing it at all.

Same problem, different terms, no translation layer



The mapping is tractable. Its absence is inertia.

10/12

rows are outright gaps in US supervision. Two are partial. None is fully addressed.

Skill atrophy (Bainbridge 1983)	→	No override-rate threshold or baseline	✗
Misuse / commission errors (Skitka 1999)	→	Effective challenge defined only qualitatively	✗
Situation awareness — projection (Endsley 1995)	→	Agentic oversight absent (fn. 3 excludes it)	✗
Human-AI complementarity (Bansal 2021)	→	No complementarity assessment required	✗

WHERE OTHERS GOT FURTHER

Three instruments already crossed the gap

JURISDICTION	INSTRUMENT	HCI CONCEPT ADDRESSED	BINDING?
EU	AI Act Article 14(4)(b)	Detect & address automation bias; operator training	Yes
Singapore	MAS FEAT + Veritas toolkit	Human-oversight assessment, co-built with major banks	Principles
International	BIS FSI Insights No. 63	'Human-in-control'; expertise for AI-managing staff	Guidance
US banking	SR 26-2	Human oversight as principle; GenAI/agentic excluded (fn. 3)	Principles

THE DEBT, ACROSS ALL THREE PARTS

The research wasn't missing. The crosswalk was.

PART 1

The ironies were named

Skill atrophy, the training paradox, opacity — predicted in 1983, reproduced across four banking waves, cited zero times in US supervision.

PART 2

The failures were measured

Forty years of automation-bias evidence shows "the analyst will catch it" is unreliable — and that the failure mode differs by automation level.

PART 3

The translation is owned by no one

Ten of twelve crosswalk rows are gaps. International regulators have begun; US banking supervision hasn't identified it as a problem.

Reproducing the ironies a fourth time is a choice — made by default, by people who haven't been told the choice exists.